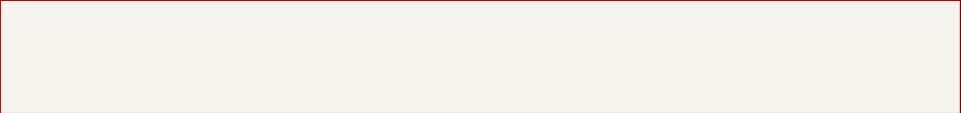


MACH3-SIMPLE – 6 axes, two LPT ports, buffered 5 V outputs

A copy of the board in service, extras removed. NOT ISOLATED on the output side.

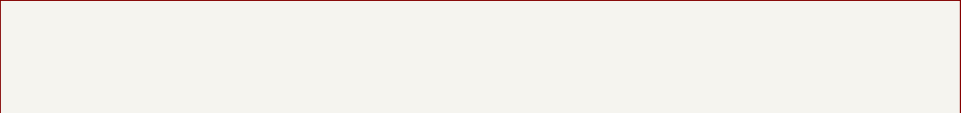
A0: NOT BUILT, NOT TESTED.

Power – 5 V and 24 V in



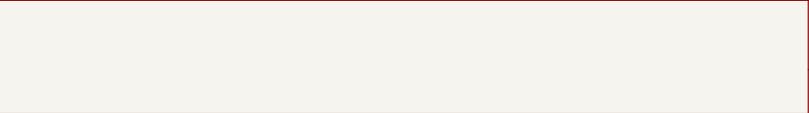
File: 01_power.kicad_sch

Inputs – home, E-stop, THC, handwheel



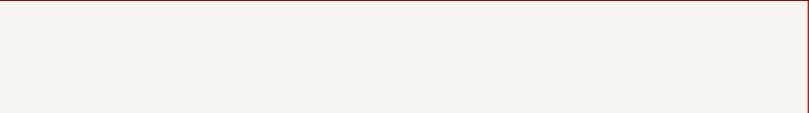
File: 03_inputs.kicad_sch

LPT port 1 – 6 axes, buffered 5 V out



File: 02_outputs.kicad_sch

RS232, MCU and six relays



File: 04_relays.kicad_sch



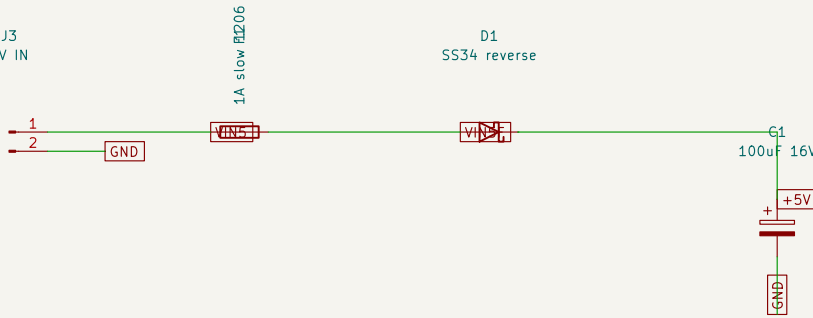
Sheet: /		
File: MACH3SIMPLE.kicad_sch		
Title: MACH3-SIMPLE: copy of the board in service		
Size: A4	Date:	Rev: A0 ENGINEERING ONLY
KiCad E.D.A. 10.0.6		Id: 1/5

POWER. Two supplies from the cabinet, exactly as the board in service.

One common ground: the outputs are not isolated, so there is nothing to keep apart. See the header of build_project.py for what that costs.

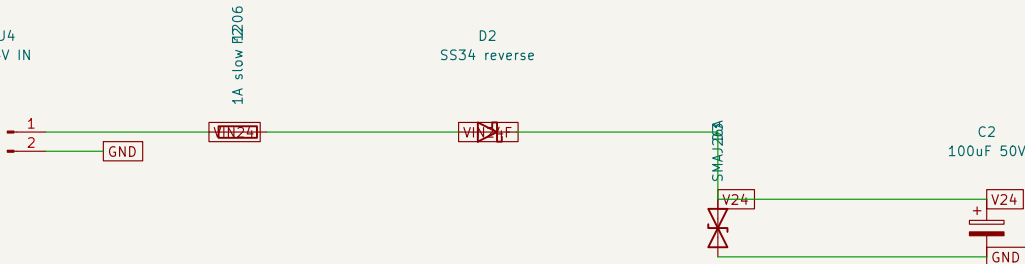
5 V LOGIC SUPPLY

Feeds the buffers, the optocoupler LEDs on the PC side, the MCU and the RS232 transceiver. F1 protects the board, not the supply.



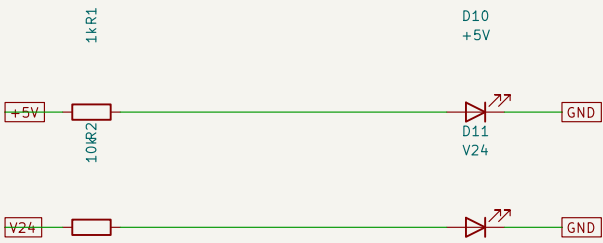
24 V SENSOR AND RELAY SUPPLY

Feeds the sensor loops and the relay coils. Nothing on this board switches mains; the relay contacts drive 24 V coils only.



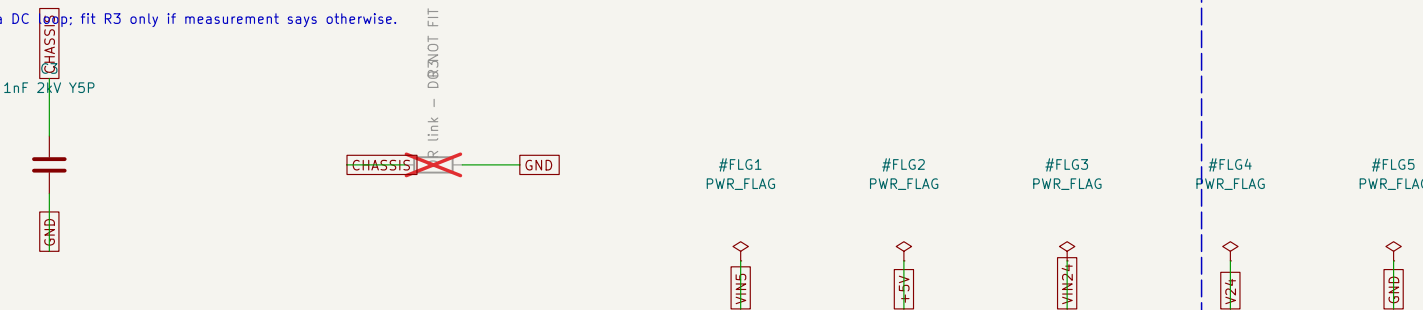
RAIL INDICATORS

See at a glance which supply is missing. 10k on 24 V, not 4k7: 4k7 would be $(24-2)^2/4700 = 103 \text{ mW}$ in a 100 mW 0603.

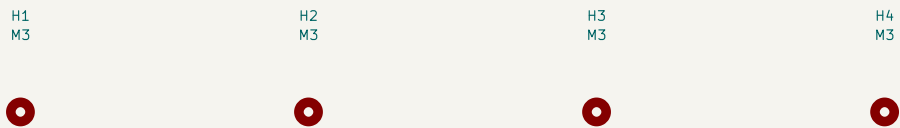


SHELL AND POWER FLAGS

The two D-sub shells share CHASSIS. C3 returns high-frequency screen current without a DC drop; fit R3 only if measurement says otherwise.



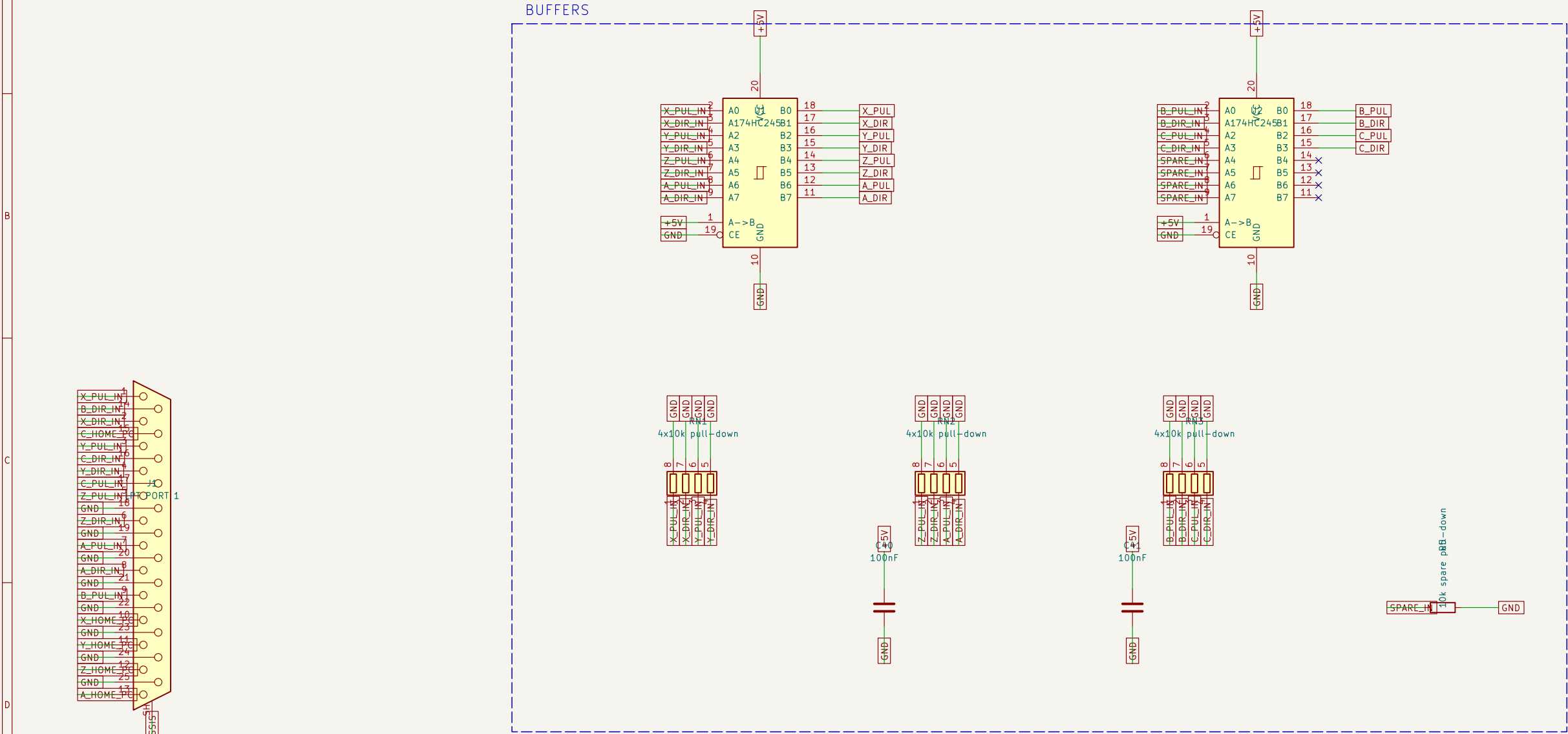
MOUNTING



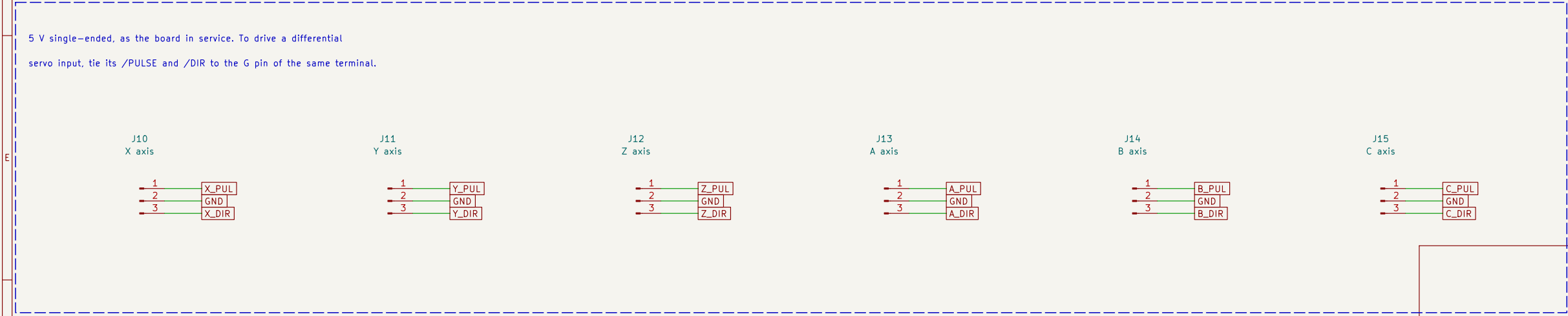
LPT PORT 1 – MOTION. Pin numbers are the ones printed on the board in

service, so the existing Mach3 profile carries over unchanged.

NOT ISOLATED: the axis terminals share ground with the PC. That is the arrangement on the board in service and the one chosen for this copy.



AXIS TERMINALS – PULSE / GND / DIR



INPUTS. 24 V sensors on optocouplers, handwheel straight through.

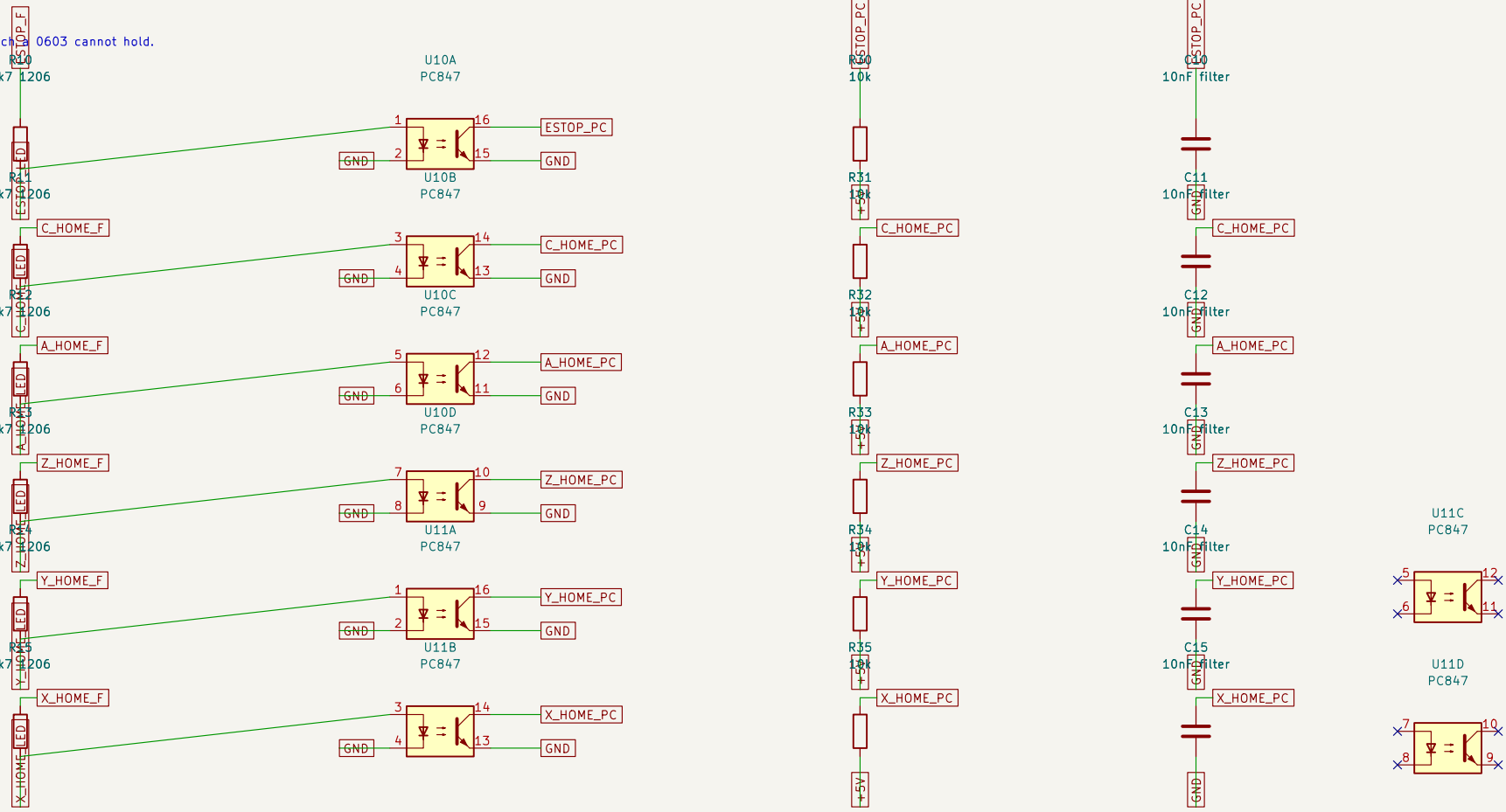
EVERY optocoupler LED gets an anti-parallel diode. A PC847 LED is rated

6 V reverse and a sensor cable beside a motor lead delivers far more.

That is the most likely reason optocouplers keep dying, and the fix is one diode each.

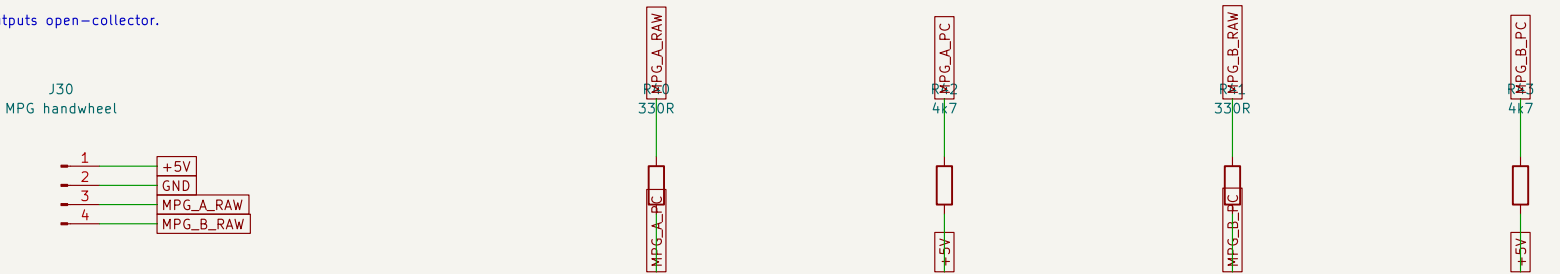
24 V INPUT CHANNELS

4k7 in 1206: $(24-1.2)^2/4700 = 0.11$ W, which 0603 cannot hold.



HANDWHEEL — 5 V, STRAIGHT THROUGH

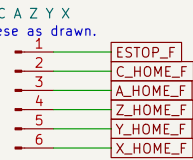
The MPG runs at 5 V and shares ground with the port, so it needs no optocoupler — only a series resistor and a pull-up, because most handwheels drive their outputs open-collector.



EMG & HOME — ONE POSITION PER SIGNAL

One six-way block, labelled EMG C A Z Y X as on the board in service.

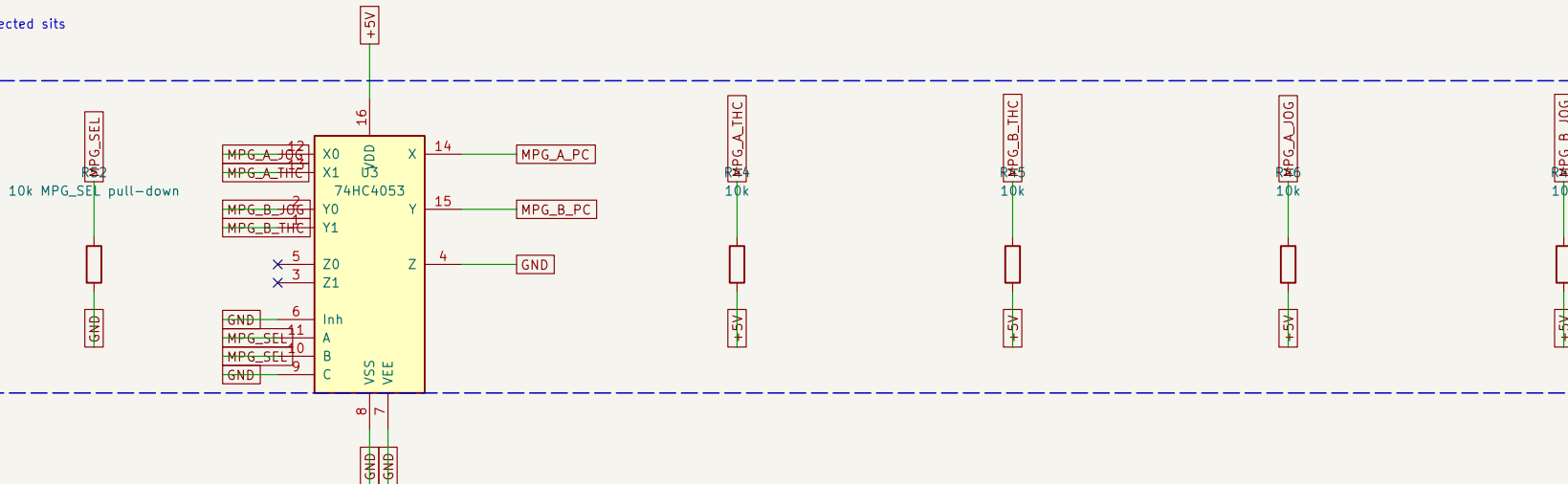
The sensors take their +24 and 0 V from the 24 V terminal, not from a 24 V terminal. The sensors take their +24 and 0 V from the 24 V terminal, not from a 24 V terminal.



HANDWHEEL ROUTING — JOG OR THC, CHOSEN BY SendSerial

One encoder, two jobs. U3 sends MPG A and B either to port 2 pins 12 and 13, where Mach3 reads them as the handwheel, or to pins 10 and 11, where it reads them as THC up and down. The MCU picks, on SendSerial 'T' for THC and 't' for jog, so it is one more macro alongside the relays.

The four LPT pins are pulled up, so whichever pair is not selected sits high and inactive rather than floating.



RS232 RELAY BANK. SendSerial("A").."F" on, "a".."f" off – the same

protocol as the board in service, so existing macros keep working.

RS232 AND MCU

The ATmega arrives blank: program it through J60 with a USBasp (firmware/build.sh flash), which also sets the fuses for the 16 MHz crystal Y1: low FF, high D9, extended FD. The factory fuses select the internal RC, which is trimmed only to 10 percent. Serial is 9600 8N1.

SIX RELAYS – 24 V COILS ONLY

The contacts switch 24 V Finder coils. Nothing on this board touches mains. A Finder coil is 20–60 mA, so a 10 A contact is heavily derated.

